Frequentist Hyp. Tests: 1) No effect is a strawman

Reject null if $p<0.05$

2) Binary

3) Conflated

→ Think about what is "substantively significant"

→ Always exists a sample size to cause to reject H_0

Frameworks

1) Frequentist: → Single value of θ (NOT an RV)

→ estimate uncertainty of sampling

2) Bayesian: → θ = RV

→ trying to estimate dist. of likely values of θ

→ explicitly account for prior beliefs

1) Hyp. Testing: → Frequentist

→ Binary decision

2) MLE: → Frequentist

→ "Most likely" value estimate

→ Gives samp. dist. of estimate, asymptotically $p(\text{data}|\theta)$

3) Bayesian: → estimate $p(\theta|\text{data})$

→ intuitive

Confidence Intervals

→ MLE : $\hat{\theta} = \theta$, $\hat{\theta} \sim N(\theta, \sigma^2)$

→ Bernstein-von Mises Theorem: dist. of MLE $p(\text{data}|\theta)$ converges to Bayesian posterior $p(\theta|\text{data})$ as $n \to \infty$ given "reasonable" priors

↳ MLE samp. dist. converges when priors are weak

Experiments

- Issues:

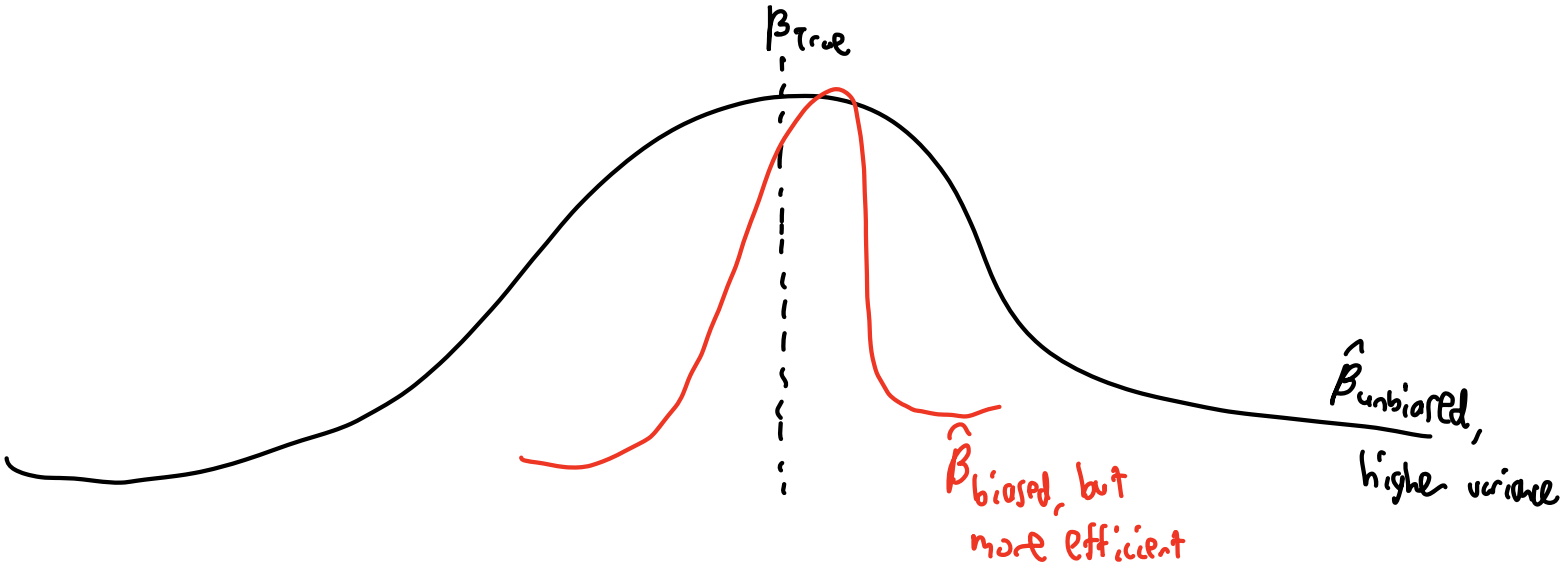
1) No baseline differences: likely given randomization + Law of Large Numbers
2) No differential treatment effects
3) SUTVA / No interference: critical reasoning / domain expertise

Internal Validity Checks

1) Balance Tests: pre-treatment covariates balanced
2) Sample Ratio: # ppl in treatment + control matches expectation given randomization
   (t-test: pick p-value which you would reject)
3) A/A Test: checks whether testing infrastructure induces differences
   ↳ same treatment for different groups => different results

Hierarchical vs. Fixed

- Difference: add extra assumption to fixed effects => hierarchical
  - ↳ fixed: less bias
  - ↳ Hierarchical: random effects, differences are uncorrelated over time, variance ↓ => better estimates?



→ better? => depends on what you want

→ clustering SE => ↑ p-value
  - ↳ assuming ind. obs.
  - ↳ adjusts SEs => less info in one person for 4 diff't features than 4 people for one feature

↳ ≈ smaller sample size

Functional Forms

- FF assumptions
- Promise of Matching helps solve problems:

① Mahalanobis: $\sqrt{\left(\frac{x_2 - x_1}{\sigma_x}\right)^2 + \left(\frac{y_2 - y_1}{\sigma_y}\right)^2}$

→ Pairing treatment + control => similar potential outcomes
↳ Norm. by σ => impact on potential outcomes

- Matching does not assume FFs like regressions
↳ have to define similarity measure
↳ choose feature weights

- Use SRS
↳ Matched pairs = limited case

CoD, PSM, Matching

02/24

→ Ex. 1: Coarse Exact Matching on 200 obs. 100 T, 100 C,
2 covariates: $x_1, x_2 \in 1,2,3,4$

How many T, C in each bucket?
↳ $\frac{100}{4^2} = \frac{100}{16} = 6.25$ in T, 6.25 in C

→ Ex. 2: Ex. 1 + 3 covariates
↳ $\frac{100}{4^3} = \frac{100}{64} = 1.5$ T, 1.5 C

→ Ex. 3: 4 covariates
↳ $\frac{100}{4^4} = \frac{100}{256} = 0.47, 0.4C$

overlap, hard to distinguish

→ CoD: adding dims ↑ distances ⇔ pts. far => challenge matching covariates
↳ Allure of Propensity Score Matching

- Conditional Independence Assumption
  - ↳ no baseline differences:
    +
    no differential treatment effects

$$(Y^1, Y^0) \perp\!\!\!\perp D \mid X$$

if $(Y^1, Y^0) \perp\!\!\!\perp D \mid X$,
then $(Y_i^1, Y_i^0) \perp\!\!\!\perp D_i \mid p(X_i)$

$p(X_i)$ = propensity score
$p(X) \equiv p_r(D_i = 1 \mid X_i)$

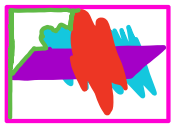
→ PSM:
↳ why does matching on $p_r(D=1 \mid X_i)$ work?
p(obs in treatment)

↳ baseline differences violated when X_i correlated w/ Y_i^0 and D_i
↳ PSM just focuses on breaking correlation ↔ X_i and D_i

→ Strengths: ① Matching on scalar easy
② Breaks correlation ↔ X_i, D_i

→ Weaknesses: ① Balance can go down, var ↑
↳ entails randomly pairing obs when $P \simeq 0$
② Assumes known $p_r(D_i = 1 \mid X_i)$ → issues when estimating

→ DAME-FLAME package



→ When can't randomize, can't rely on LLN => obs. data

→ Suppose there are baseline differences

↳ Adjusting: $E[Y^0 | D=1, \underbrace{X}_{\text{not explicitly observable}}] = E[Y^0 | D=0, X]$

- Omitted Variable Bias: effect of D_i on Y

↳ Suppose $Y_i = \alpha^L + \beta^L D_i + \gamma A_i + e_i^L$

→ Want to estimate: $Y_i = \alpha^S + \beta^S D_i + e_i^S$

OVB = difference ↔ β^L and β^S

= relation ↔ A_i, D_i • effect of A_i

$\Rightarrow A_i = \pi_0 + \pi_1 D_i + \mu_i = \pi_i$

→ Power ≈ 80%

→ OEC : 1) Measurable

2) Successful

3) Sensitive : can be moved enough by experiment to detect it, but must be short-term responsive

(4) Repeated searches over short term)

Endogenous Stopping (Peeking)

$p(H_0 = \text{true} \mid \hat{y}) = \text{unknown}$

$\text{p-value} \neq p(H_0 = \text{true} \mid \hat{y})$

$\text{p-value} = p(\hat{y} \mid H_0 = \text{true})$

→ By Bayes: $p(H_0 = \text{true} \mid \hat{y}) = \dfrac{p(\hat{y} \mid H_0 = \text{true}) \cdot p(H_0 = \text{true})}{p(\hat{y})}$

Linear Probability Models

→ variance of errors

→ Heteroshedasticity (H3)

↳ standard errors do not behave well w/ n low

→ LPM : predicted outside (0,1)

↳ constant marginal effects

↳ Heteroshedastic SE : robust SE = correction factor (var in errors cannot be correlated w/ $X\beta$)

→ Logit : predict $\in [0,1]$

↳ variable marginal effects

↳ fine for small samples (OK for any sample, assuming model is correct)

→ p-values:

↳ model uncertainty

↳ sampling uncertainty

2/05/2026

Power

→ Attempt to tolerate 5% chance Type I error,
20% " " II "

Stable Unit Treatment Value Assumption

→ Ex. Covid vax given to half of population

↳ Why violating SUTVA?

→ Treatment status of individual depends on treatment rates of population

→ Estimated effect of vax will be lower than true ATE

↳ How to fix issue?

→ Treat entire dorms / villages?

→ Ex. Uber drivers get more per trip → 10% drivers earn more

↳ Violation

→ May incentivize drivers to get on road more

→ Makes more difficult to pick up riders

→ Treatment effect ↑

Ø General Equilibrium Effect

↳ How to fix issue

→ Assign entire cities control or treatment

→ Randomize across days/weeks

→ Ex. Instagram feed shows happy content for 10%

↳ Violation

→ Content sharing

→ Fix?

→ Temporal Separation

1/22 - Potential Outcomes

→ Heterogeneous Treatment Effects are always present

↳ Ex. Migraines

→ Attenuation Bias: Coefficient tends toward zero

$T \in \{0,1\}$

$y_i^{T=0} \equiv y_i^0$

$y_i^{T=1} \equiv y_i^1$

$\delta_i = y_i^1 - y_i^0$ → difference ↔ 2 worlds

for any population, regardless of world

$\frac{\sum \delta_i}{N} \equiv$ ATE => average treatment effect → average counter-factual over everybody

↳ $E[Y^1] - E[Y^0]$

estimatable quantity for value in individual unchanging world

Can measure: ① $E[Y^1 | D=1] - E[Y^0 | D=0] = \widehat{ATE}$

D = obs. treatment status of individuals

↳ fixed quantity

for $ATE = \widehat{ATE}$: when is estimate equal to obs.

↳ $E[Y^1|D=1] - E[Y^0|D=0] + E[Y^0|D=1] - E[Y^0|D=1]$

=> $\underbrace{E[Y^1|D=1] - E[Y^0|D=1]}_{\text{Average Treatment on Treated (ATT)}} + \underbrace{E[Y^0|D=1] - E[Y^0|D=0]}_{\text{baseline differences: difference in untreated world}}$

↳ counterfactuals of each other

Assume $\sum$ Baseline Differences $= 0$: ATT $\stackrel{?}{=}$ ATE

$ATE = \lambda \underbrace{(E[Y^1|D=1] - E[Y^0|D=1])}_{ATT} + (1-\lambda) \underbrace{(E[Y^1|D=0] - E[Y^0|D=0])}_{ATT^c}$

$\lambda = \frac{\text{size } D=1}{\text{size } N}$ = prop. of population that's treated

T (states of world)

D (population)	0	1
0	63 ①	50 ②
1	63 ④	50 ③

1) What squares are observable: ①, ④

2) No baseline differences implies what squares are equal? => ①, ③

Corr w/ D AND Y^0, i.e. confounding (potentially gender)

3) If $ATT = ATT^c$, what equation is true? => ③ - ④ = ② - ①

4) What if ④ = 50? => Fitbit may not have a significant effect on heart rate

ATE: Considering taking actions at scale

Key Questions: ① No Baseline Differences

↳ NOT required: no differences ↔ D=0, D=1

↳ " " : no differences in Y across individuals

↳ Requires: no average differences in outcome Y ↔ D=0, D=1

② No Differential Treatment Effects

↳ observable effects?

↳ effect of T is same on avg. for both groups

↳ if not met, but no baseline differences is met, still get treatment effect $\boxed{\text{ATT}}$

- Time - staggered: treatments assigned at diff't times => convergence
- External Validity: effect of study generalizes to larger/diff't context

① Entities drawn from same pop? → sample composition
↳ Ex. White men aftelimes studied in health-based studies

② Treatment same in new setting?
↳ Car ex. Generalize to new, smaller discounts

③ Broader context in new setting?
↳ More control => ↑ internal validity => ↑ artificiality
↳ Hawthorne Effect: → Ex. Social desirability

EV defined w.r.t. context to which are willing to generalize the results

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→ Ex. 1: DHCC has new hard cider, sales are weak → give free samples at restaurants => sales jump 30%
→ EV of a study is not singular

→ Ex. 2a: Apple $100 discount on laptops
2b: Car $1K discount used car site
→ Primacy + Novelty
→ Sales + promotions => ① Win customers from competitors (Real gains)
② Time-shift purchases (Illusory gains)

→ Ex. 3a: Bing new worse algorithm
3b: Google " " "

→ Ex. 4: Google providing recipe sites
→ Adversarial Users: change behavior in response to deployment of algorithm
→ General Equilibrium Effects
→ Strategic Acting